GCSE Biology Paper 1

Triple

freesciencelessons

Cell Biology

Organisation

Infection and Response

Bioenergetics

Dr Shaun Donnelly

Introduction

When I started freesciencelessons in 2013, I had one simple goal: to help students in their understanding of science. When I was at school (and we're talking thirty years ago now), science was always my favourite subject. It's not surprising that I went on to become a science teacher. I know that many students find science challenging. But I really believe that this doesn't have to be the case. With patient teaching and a bit of hard work, any student can make amazing progress.

Back in 2013, I had no idea how big freesciencelessons would become. The channel now has over 160 million views from 192 countries with a total view time of over 300 years. I love to hear from the students who have patiently watched the videos and realised that they can do science after all, despite in many cases having little confidence in their ability. And just like in 2013, I still make all the videos myself (many students think that I have a staff of helpers, but no, it's just me).

This workbook is designed to complement the Biology 1 videos for the AQA specification. However, there is a huge amount of overlap with other exam boards and in the future I'll be making videos and workbooks for those as well. I've packed the workbook full of questions to help you with your science learning. You might decide to start at the beginning and answer every question in the book or you might prefer to dip in and out of chapters depending on what you want to learn. Either way is fine. I've also written very detailed answers for every question, again to help you really develop your understanding.

Please don't think of science as some sort of impossible mountain to climb. Yes there are some challenging bits but it's not as difficult as people think. Take your time, work hard and believe in yourself. When you find a topic difficult, don't give up. Just go to a different topic and come back to it later.

Finally, if you have any feedback on the workbooks, you're welcome to let me know (support@freesciencelessons.co.uk). I'm always keen to make the workbooks better so if you have a suggestion, I'd love to hear it.

Good luck on your journey. I hope that you get the grades that you want.

Shaun Donnelly

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Revision Tips

The first important point about revision is that you need to be realistic about the amount of work that you need to do. Essentially you have to learn two years of work (or three if you start GCSEs in Year 9). That's a lot of stuff to learn. So give yourself plenty of time. If you're very serious about getting a top grade then I would recommend starting your revision as early as you can. I see a lot of messages on Youtube and Twitter from students who leave their revision until the last minute. That's their choice but I don't think it's a good way to get the best grades.

To revise successfully for any subject (but I believe particularly for science), you have to really get into it. You have to get your mind deep into the subject. That's because science has some difficult concepts that require thought and concentration. So you're right in the middle of that challenging topic and your phone pings. Your friend has sent you a message about something that he saw on Netflix. You reply and start revising again. Another message appears. This is from a different friend who has a meme they want to share. And so on and so on.

What I'm trying to tell you is that successful revision requires isolation. You need to shut yourself away from distractions and that includes your phone. Nothing that any of your friends have to say is so critically important that it cannot wait until you have finished. Just because your friends are bored does not mean that your revision has to suffer. Again, it's about you taking control.

Remember to give yourself breaks every now and then. You'll know when it's time. I don't agree with people who say you need a break every fifteen minutes (or whatever). Everyone is different and you might find that your work is going so well that you don't need a break. In that case don't take one. If you're taking breaks every ten minutes then the question I would ask is do you need them? Or are you trying to avoid work?

There are many different ways to revise and you have to find what works for you. I believe that active revision is the most effective. I know that many students like to copy out detailed notes (often from my videos). Personally, I don't believe that this is a great way to revise since it's not really active. A better way is to watch a video and then try to answer the questions from this book. If you can't, then you might want to watch the video again (or look carefully at the answers to check the part that you struggled with).

The human brain learns by repetition. So the more times that you go over a concept, the more fixed it will become in your brain. That's why revision needs so much time because you really need to go over everything more than once (ideally several times) before the exam.

Revision Tips

I find with my students that flashcards are a great way to learn facts. Again, that's because the brain learns by repetition. My students write a question on one side and the answer on the other. They then practise them until they've memorised the answer. I always advise them to start by memorising five cards and then gradually adding in extra cards, rather than try to memorise fifty cards at once.

I've noticed over the last few years that more students do past paper practise as a way of revising. I do not recommend this at all. A past paper is what you do AFTER you have revised. Imagine that you are trying to learn to play the guitar. So you buy a guitar and rather than having lessons, you book yourself into a concert hall to give a performance. And you keep giving performances until you can play. Would you recommend that as a good strategy? I wouldn't. But essentially that's how lots of students try to revise. Yes by all means do practise papers (I've included a specimen paper in this book for you) but do them at the end when you've done all your revision. Past papers require you to pull lots of different bits of the specification together, so you should only do them when you are capable of that (ie when you've already done loads of revision).

A couple of final points ....

There will be times when I decide to update a book, for example to make something clearer or maybe to correct a problem (I hope not many of those). So please keep an eye out for updates. I'll post them on Twitter (@UKscienceguy) and also on the FAQ page of my website. If you think that you've spotted a mistake or a problem, please feel free to contact me.

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Chapter 1: Cell Biology

• Describe what is meant by a prokaryotic and eukaryotic cell and describe the features of these cells.
• Use common size units in standard and non-standard form and size prefixes (eg milli, micro, nano etc).
• Make order of magnitude calculations.
• Describe the main structures in animal cells and the functions of these structures.
• Describe the main structures in plant cells and the functions of these structures.
• Describe how different animal cells can be specialised to carry out their functions.
• Describe how different plant cells can be specialised to carry out their functions.
• Describe how to use a light microscope to view and draw plant and animal cells (required practical).
• Describe the advantages of electron microscopes over light microscopes.
• Make calculations involving magnification.
• Describe how bacteria divide by binary fission and calculate the number of bacteria present after a certain period of time has elapsed.
• Describe how to prepare a bacterial agar plate and investigate the effects of antibiotics or antiseptics on bacterial growth (required practical).
• Describe the process of cell division by mitosis and the role of cell division by mitosis in living organisms.
• Describe the features of stem cells (both embryonic and adult) and how stem cells can be used in medicine (including the role of embryonic cloning).
• Describe how particles can move by diffusion and describe the factors that affect the rate of diffusion.
• Calculate the surface area : volume ratio for an organism and explain how gills increase the surface area for diffusion of gases in fish.
• Describe what is meant by osmosis.
• Describe how to investigate the effects of osmosis on plant tissue (required practical).
• Describe how particles can move by active transport and describe the differences between active transport and diffusion.

Plant Cell Specialisation

1. The structures found in eukaryotic cells are listed below.

Circle the structures that are found only in plant cells.

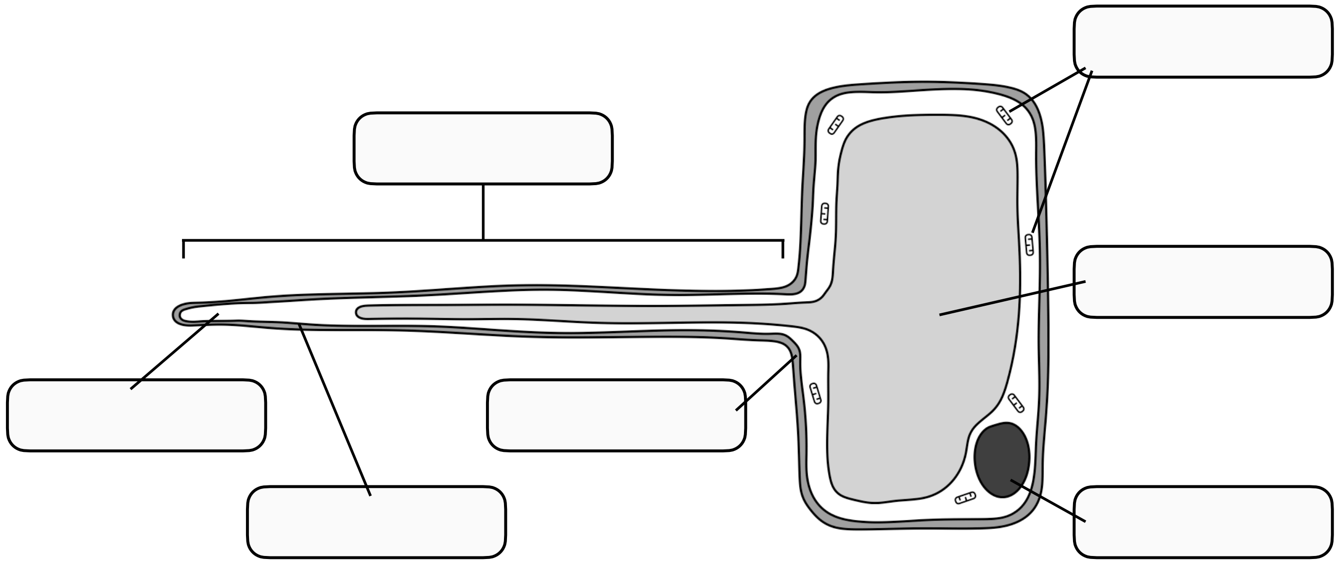
Cell membrane	Vacuole	Cytoplasm	Chloroplasts
Mitochondria	Ribosomes	Cell wall	Nucleus

2. Root hair cells are found in the roots of plants.

a. State the function of a root hair cell.

b. A diagram of a root hair cell is shown below.

Label the diagram to show the structures found in a root hair cell.

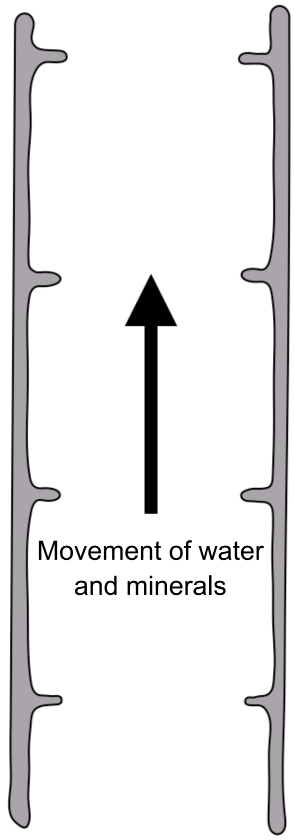


c. Explain how the root hair allows the cell to carry out its function.

d. Why do root hair cells not contain any chloroplasts?

3. Another type of specialised plant cell is found in the xylem. Xylem form hollow tubes running through the stem of a plant.

a. Describe the function of xylem.



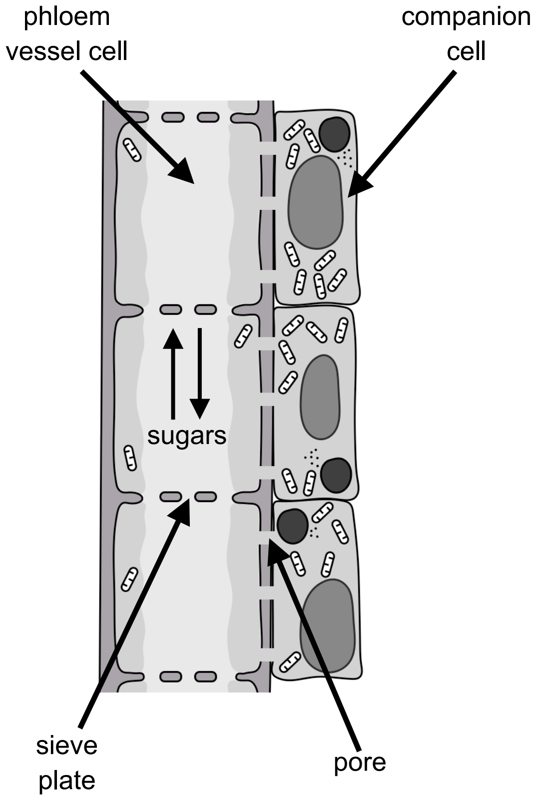
b. The diagram shows a xylem vessel.

Label the diagram to show the thick walls containing lignin and the remains of end walls.

c. Complete the table to show the purpose of the different adaptations found in xylem.

Adaptation	Purpose
Thick walls containing lignin	
No end walls between the cells	
No internal structures eg nucleus	

4. Phloem tubes are also found in the stem of plants. The diagram shows part of a phloem tube.



a. What is the function of phloem tubes?

b. How are the phloem vessel cells adapted to carry out this function?

c. What is the role of the companion cells?

Required Practical: Effect of Osmosis on Plant Tissue

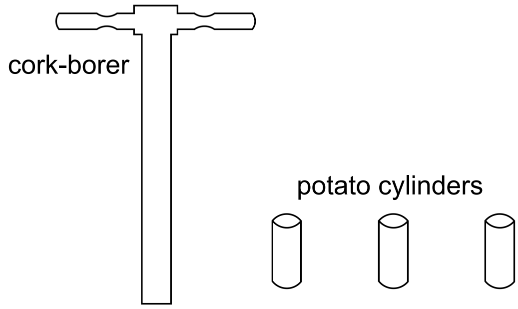
Exam tip: Like all required practicals, try to get an overall idea of the equipment and method. Do not get hung up on exact measurements or quantities.

1. In this practical we are looking at the effect of osmosis on potato cells. Any other root vegetable also works well.

a. We start by peeling the potato. Explain why we need to remove the potato skin.

b. We then use a cork-borer to produce cylinders of potato.

Explain the advantage of using a cork-borer.



c. We then use a scalpel to cut all of the potato cylinders to the same length (eg 3 cm).

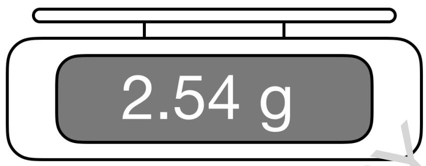
Why is it important that the cylinders are not too short?

d. Both the diameter and length of the cylinders are examples of (circle the correct answer):

Independent variables | Control variables | Dependent variables

2. Next we measure the mass and the length of the potato cylinders.

The resolution of a measuring instrument is the smallest value that can be measured. State the resolution of the balance on the right.



3. We now place the cylinders in different concentrations of sugar solution and leave overnight.

What do you think is the advantage of leaving the cylinders overnight rather than for an hour?

4. Lastly, we gently roll the cylinders on paper towel and then weigh them again and measure their length again.

Why is it important that we do not press on the potato cylinders?

5. Once we have weighed and measured the potato cylinders, we need to calculate the percentage change.

a. Calculate the percentage mass change of the potato cylinders shown.

Remember to include a positive (+) sign if the cylinder gained mass and a negative (-) sign if they lost mass.

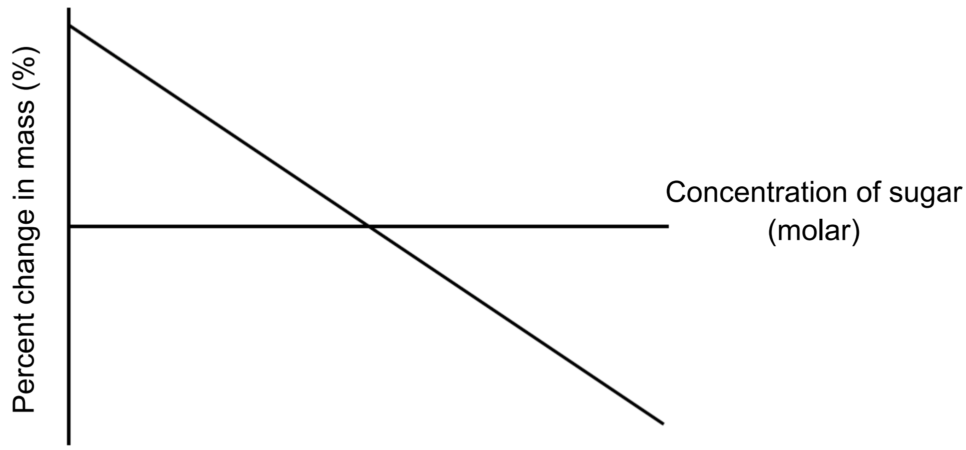
Potato cylinder 1		Potato cylinder 2	
Start mass (g)	2.75	Start mass (g)	2.78
Mass gained (g)	0.35	Mass lost (g)	0.64
% change in mass		% change in mass	

Potato cylinder 3		Potato cylinder 4	
Start mass (g)	2.82	Start mass (g)	2.73
Final mass (g)	2.26	Final mass (g)	3.40
% change in mass		% change in mass	

b. What is the advantage of looking at percentage change in mass rather than simply mass?

c. We now plot the % changes in mass against the concentration of sugar.

This is shown in the graph below.



Label the graph to show:

- Where water has moved into the potato cylinders by osmosis.
- Where water has moved out of the potato cylinders by osmosis.

d. Label the graph to show where the concentration of the sugar solution is the same as the concentration of the potato cylinder.

Explain your answer.

e. Look again at your answers to question 5a.

Place a point on the line on the graph which approximately applies to each of the four potato cylinders.

Cardiovascular Diseases

1. Cardiovascular diseases are diseases of the heart and blood vessels.

a. Cardiovascular diseases are non-communicable. What is meant by this?

b. Coronary heart disease is an example of a cardiovascular disease.

Which of the following statements is true about the coronary arteries?

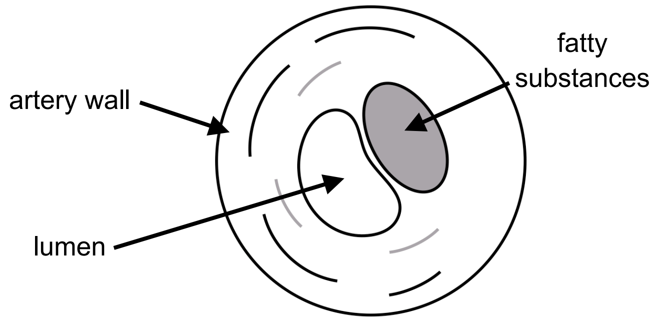
The coronary arteries branch out of the pulmonary artery and provide oxygen to the muscle cells of the heart	The coronary arteries branch out of the aorta and provide carbon dioxide to the muscle cells of the heart	The coronary arteries branch out of the aorta and provide oxygen to the muscle cells of the heart

c. Explain why the heart muscle cells cannot contract effectively if the coronary arteries are blocked.

d. The diagram shows a coronary artery in a patient with coronary heart disease.

Remember that the lumen is the space where the blood flows.

Describe what happens in coronary heart disease and explain the effect of this on the heart muscle.



e. Many patients with coronary heart disease take drugs called statins.

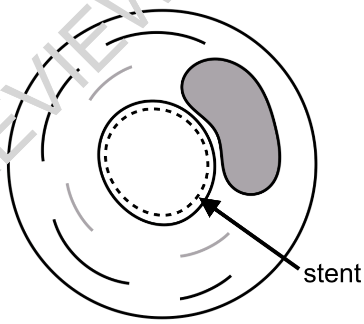
Complete the boxes below to show the positive and negative effects of taking statins.

Positive effects of taking statins	Negative effects of taking statins

f. Patients with advanced coronary heart disease can be fitted with a stent. This is shown in the diagram.

Describe how a stent works.

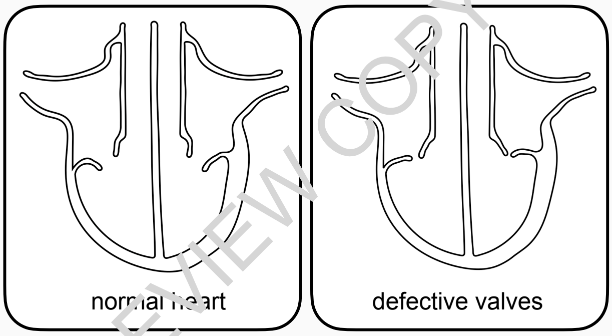
g. Describe the disadvantage of stents.



2. Some patients have problems with their heart valves. These are shown in the diagram below.

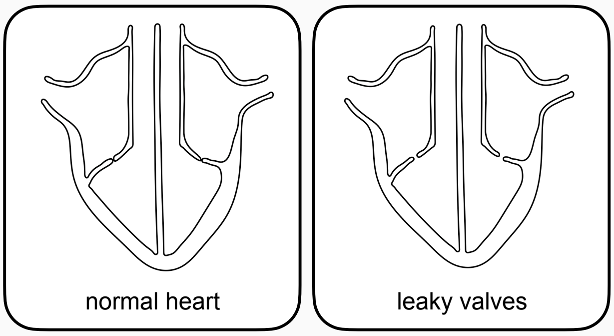
a. Beneath each diagram, describe the effect of the problem on the patient.

Heart valves do not open fully when the atria contract



Effect on patient:

Heart valves that leak blood when the ventricles contract



Effect on patient:

b. Defective heart valves can be treated using either mechanical valves or valves from animals.

Draw two lines from each type of valve to the correct description.

Mechanical valves	Valves from animals

Do not last as long and may need to be replaced	Risk of blood clotting. Patient needs to take anti-clotting drugs for rest of life	These can last a lifetime	Patients are not required to take drugs with this type of valve

3. In some cases, patients experience heart failure.

a. What is the problem with the heart in heart failure?

b. Patients with heart failure can be treated with a heart transplant or a heart-lung transplant.

Describe two problems with this.

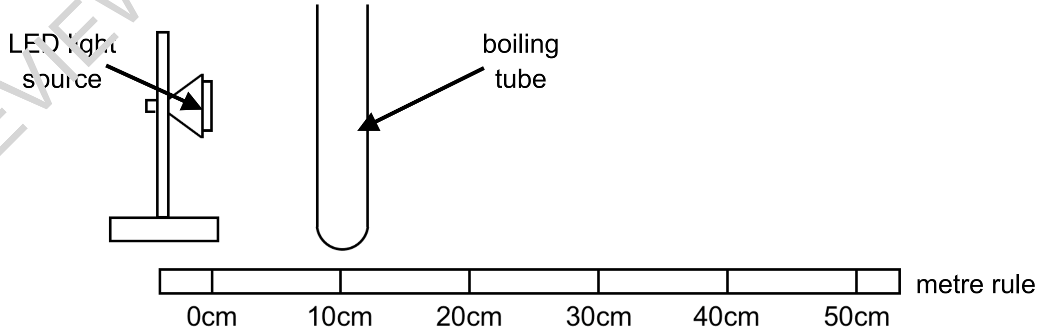
c. Some patients with heart failure can be treated with an artificial heart.

Explain why artificial hearts can only be used for a relatively short time.

Required Practical: Photosynthesis

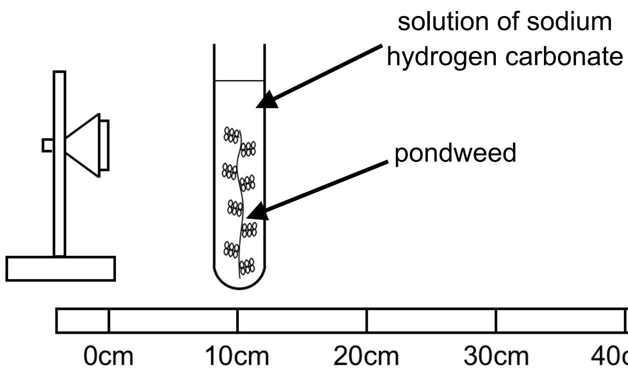
Exam tip: With any required practical, try to learn the sequence of steps.
Remember to link any step with the equipment needed for that step.

1. We start by placing a boiling tube 10 cm from an LED light source.



Explain the advantage of using an LED light source rather than a normal light bulb.

2. Now we fill the boiling tube with a solution of sodium hydrogen carbonate and place a piece of pondweed into the solution.

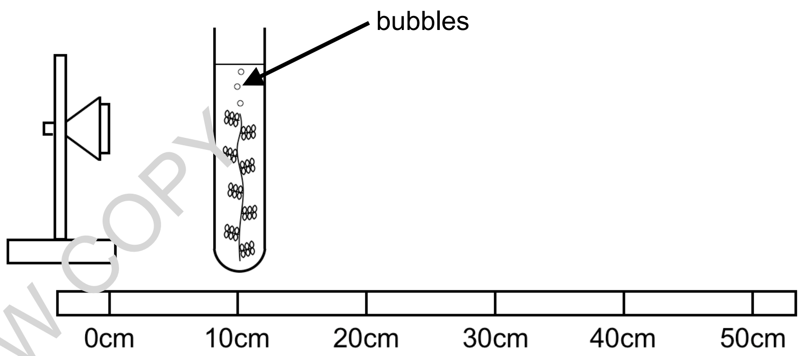


a. Explain why we use a solution of sodium hydrogen carbonate rather than simply water.

b. Why is it important that we place the pondweed with the cut side at the top?

c. We now leave the pondweed for five minutes. Why is this important?

3. Bubbles will now start to be produced from the cut end of the pondweed.



a. What is this gas and how is it produced?

b. We now start a stopwatch and count the number of bubbles produced in one minute.

We then repeat the experiment two more times and calculate a mean value.

A student's results are shown below.

Distance of pondweed from lamp (cm)	Number of bubbles produced in one minute			
	Repeat 1	Repeat 2	Repeat 3	Mean
10	36	40	12	

The results show an anomalous result. Identify the anomalous result in the table.

c. Calculate the mean value and write this in the table.

We now repeat the whole experiment increasing the distance to 20 cm, 30 cm, 40 cm and 50 cm.

4. Some of the variables in this experiment are shown below.

Write **IV** next to the independent variable, **DV** next to the dependent variable and **CV** next to the control variables.

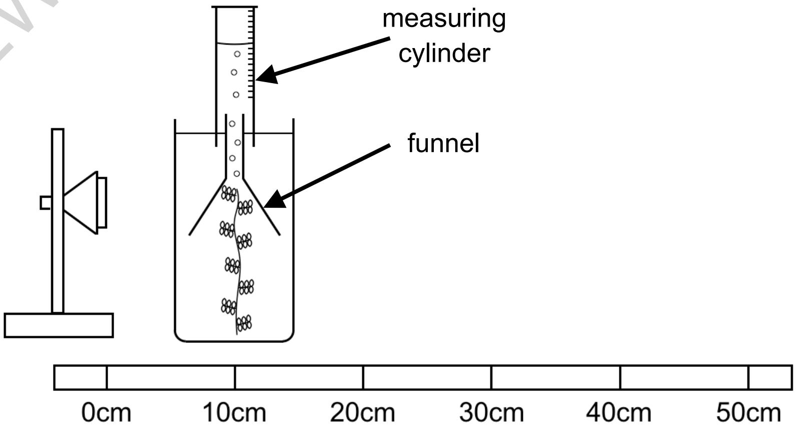
Variable	Type of variable	Reason for answer
The species of pondweed		
The concentration of sodium hydrogen carbonate solution		
The volume of sodium hydrogen carbonate solution		
The distance between the LED light and the pondweed		
The LED light source		
The temperature of the room		
The number of bubbles produced per minute		

5. There are two problems with the method outlined above.

a. Describe the two problems in the spaces below.

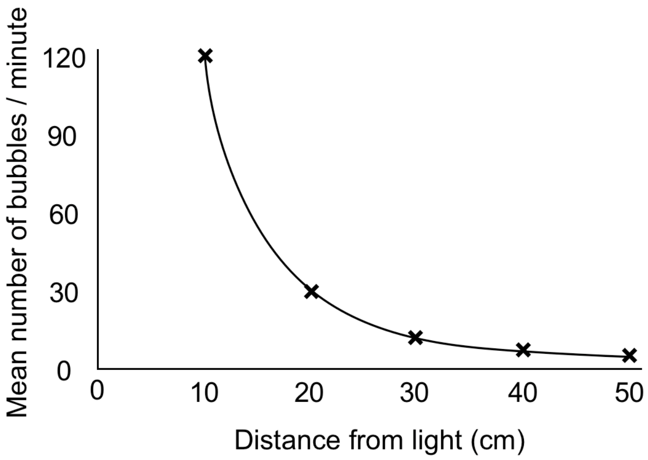
Problem 1	**Problem 2**

b. We can address these problems by using the apparatus below.



Explain how this is a more accurate way of measuring the rate of photosynthesis than the previous method.

6. The graph shows how the number of bubbles per minutes varies with the distance from the light source.



a. Use the graph to determine the mean number of bubbles per minute at the following distances.

Distance (cm)	Mean number of bubbles per minute
10	
20	
40	

b. These results follow the inverse-square law. How do the data show that?

c. Explain these results in terms of light intensity and photosynthesis.

Biology Paper 1

GCSE Specimen Paper

Time allowed: 105 minutes

Maximum marks: 100

Please note that this is a specimen exam paper written by freesciencelessons. The questions are meant to reflect the style of questions that you might see in your GCSE Biology exam.

Neither the exam paper nor the mark scheme have been endorsed by any exam board. The answers are my best estimates of what would be accepted but I cannot guarantee that this would be the case. I do not offer any guarantee that the level you achieve in this specimen paper is the level that you will achieve in the real exam.

1 Lung cancer is a non-communicable disease.

1.1 State what is meant by a non-communicable disease. **1 mark**

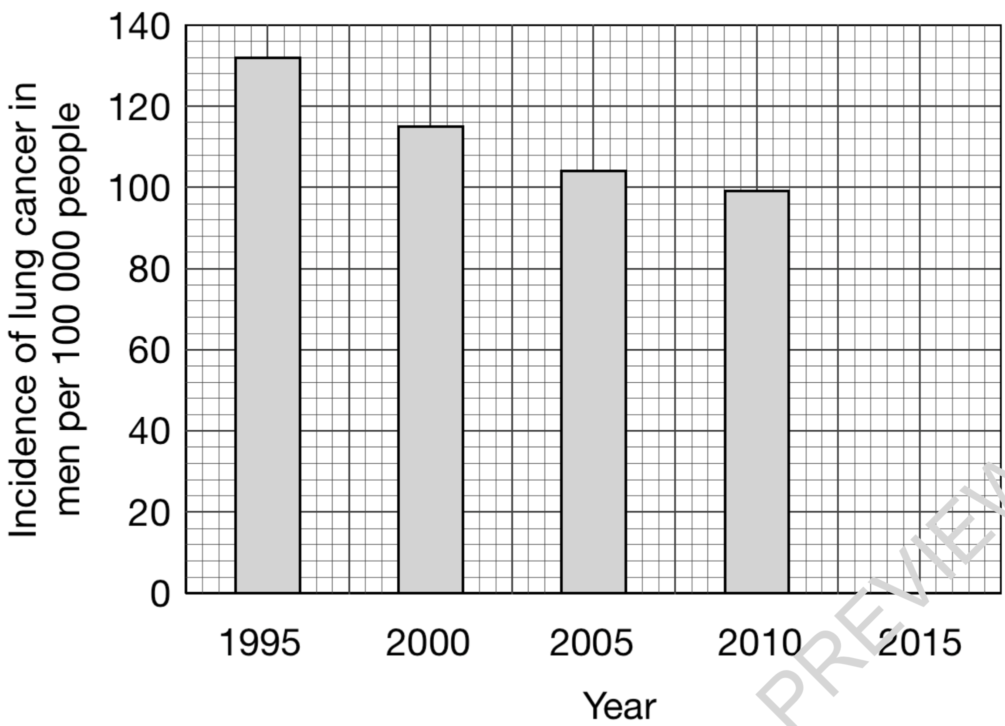
Table 1 shows how the incidence of lung cancer in men in the UK has changed since 1995.

Table 1

Year	Incidence of lung cancer in men per 100 000 people
1995	132
2000	115
2005	104
2010	99
2015	92

The data above are shown in graphical form in **Figure 1**.

Figure 1



1 . 2 Complete **Figure 1** to show the data for 2015.

1 mark

1 . 3 A student made the following statement:

"The data show that the incidence of lung cancer is falling."

Give one reason why this is **not** supported by the data in **Figure 1**.

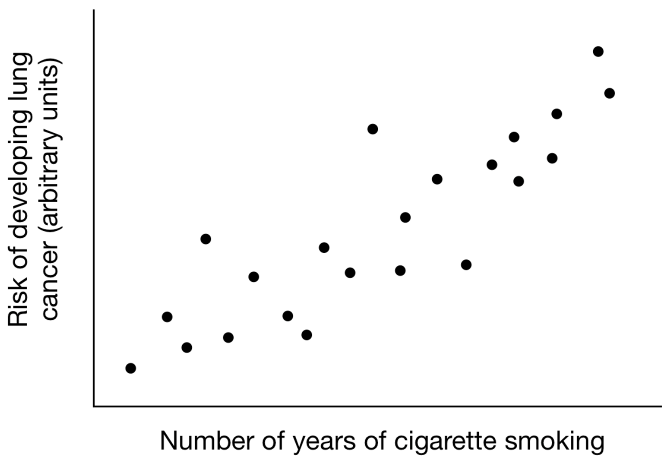
1 mark

To try to identify the cause of lung cancer, scientists carried out sampling.

The scientists asked a number of people about their smoking habits.

This produced a scattergraph similar to that shown in **Figure 2**.

Figure 2



1 . 4 Explain how the scattergraph in **Figure 2** suggests that the risk of developing lung cancer is correlated to the number of years a person smokes cigarettes.

1 mark

1 . 5 Explain how the data do not prove that smoking cigarettes causes lung cancer.

1 mark

1 . 6 Why is it important that sampling involves a very large number of people?

1 mark

____

____

1 . 7 Scientists have identified a number of chemicals in cigarette smoke which can increase the risk of developing cancer.

What name do scientists give to chemicals such as these?

1 mark

____

1 . 8 Describe the stages in the formation of a cancer in humans.

4 marks

____

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____

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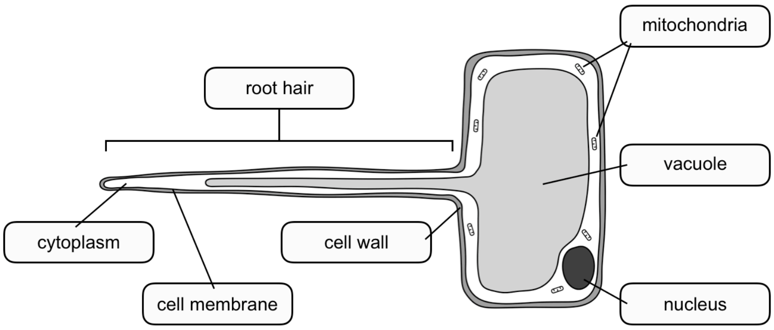
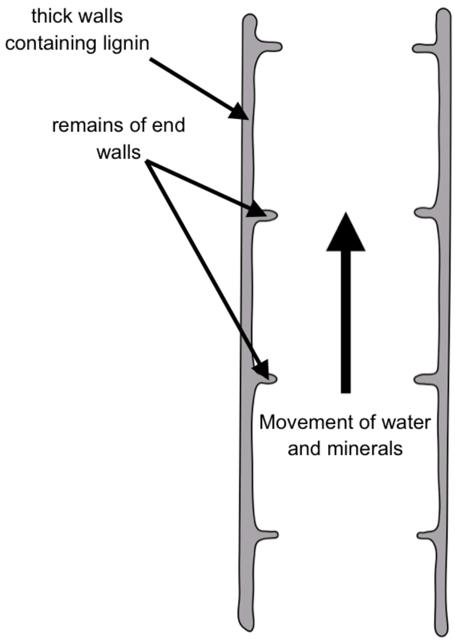
Total = 11

Biology Paper 1
Triple

Answers to Questions

Plant Cell Specialisation	
1	The vacuole, chloroplast and cell wall are only found in plant cells. I should point out that in some textbooks, you might see the idea of vacuoles in animal cells. This is correct. Animal cells do have temporary, short-lived vacuoles. However, we only find a permanent vacuole in plant cells and the AQA specification only requires you to know about these.
2a	The function of a root hair cell is to absorb water and minerals into the plant root.
2b	
2c	The root hair increases the surface area of the cell, allowing it to absorb water and dissolved minerals more effectively.
2d	Chloroplasts use light energy to carry out photosynthesis. The root hair cells are found underground, where there is no light. This means that root hair cells do not have chloroplasts.
3a	The xylem carries water and dissolved minerals up the stem from the roots to the leaves.
3b	
3c	(see table below)

Adaptation	Purpose
Thick walls containing lignin	These provide support to the plant
No end walls between the cells	The xylem cells form a long, hollow tube allowing water and minerals to flow more easily
No internal structures eg nucleus	This makes it easier for water and minerals to flow through the xylem

Plant Cell Specialisation	
4a	Phloem tubes transport dissolved sugars up and down the plant stem.
4b	Phloem vessel cells have no nucleus and very limited cytoplasm. The end walls have pores called sieve plates. Both of these adaptations allow dissolved sugars to pass easily through the phloem.
4c	Phloem vessel cells have very few mitochondria. However, transport of sugars up and down the phloem requires energy. The companion cells contain a large number of mitochondria. These provide the energy that the phloem vessel cells require.

Required Practical 3: Effect of Osmosis on Plant Tissue	
1a	The skin on a potato may interfere with osmosis, preventing water from moving in or out of the tissue inside the potato. Therefore it is important that we peel the potato before we start.
1b	When we use a cork borer, this produces potato cylinders all of the same diameter. This is important as the diameter of the potato cylinder could affect the rate of osmosis.
1c	We are going to measure the mass change or length change of the potato cylinders when they have been left in different concentrations of sugar solution overnight. Some cylinders will gain mass, as water moves in by osmosis and some cylinders will lose mass, as water moves out by osmosis. However, if the potato cylinders are too short, then the amount of osmosis that takes place may be very small and the mass / length of the cylinders may not change enough to measure. This is why we need to make sure that the potato cylinders are not too short.
1d	Both the diameter and length of the potato cylinders are examples of control variables. Control variables are variables that we keep the same in our experiment. We must make certain that the diameter and length of the potato cylinders are the same. If they are not, then this could affect how much osmosis changes the mass / length of the cylinders.
2	Remember that the resolution is the smallest value that can be measured by an instrument. In this case, we can see that the resolution of the balance shown is 0.01 g.
3	In this practical, osmosis will cause water to move in or out of the potato cylinders (depending on the concentration of sugar solution that the cylinders are in). Osmosis takes time to happen. If we left the cylinders in the solution for only one hour, then this would not be enough time for osmosis to fully take place. Because of this, the potato cylinders might not change their mass / length enough to measure. This is why we leave them in the sugar solution overnight.
4	When we take the potato cylinders out of the sugar solution, the cylinders will be wet. If we weighed them at this stage, then their masses would include the water on the surface. To remove this, we gently roll the cylinders across paper towel. However, if we pressed on the potato cylinders, then we could force water out of the potato cells and this would give us inaccurate mass changes.

Required Practical 3: Effect of Osmosis on Plant Tissue

5a Potato cylinder 1. Percentage mass change = (0.35 / 2.75) x 100 = +12.7% (to 1 decimal place). Note that this value is positive as the potato cylinder gained mass.

Potato cylinder 2. Percentage mass change = (0.64 / 2.78) x 100 = -23.0% (to 1 decimal place). Note that this value is negative as the potato cylinder lost mass.

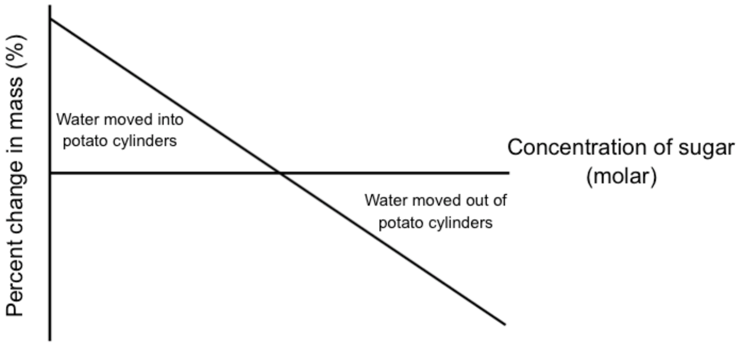
For potato cylinders 3 and 4, you have to calculate the mass change before you can use this to calculate the percentage mass change.

Potato cylinder 3. Mass change = 2.82 - 2.26 = 0.56 g mass lost.
Percentage mass change = (0.56 / 2.82) x 100 = -19.9% (to 1 decimal place). Note that this value is negative as the potato cylinder lost mass.

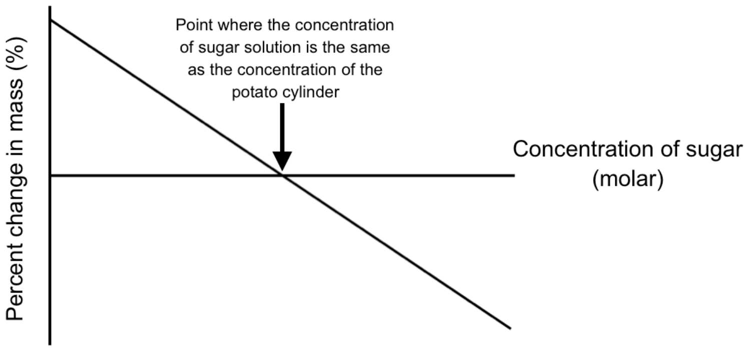
Potato cylinder 4. Mass change = 3.40 - 2.73 = 0.67 g mass gained.
Percentage mass change = (0.67 / 2.73) x 100 = +24.5% (to 1 decimal place). Note that this value is positive as the potato cylinder gained mass.

5b Percentage change in mass takes into account the different start masses of the potato cylinders.

5c

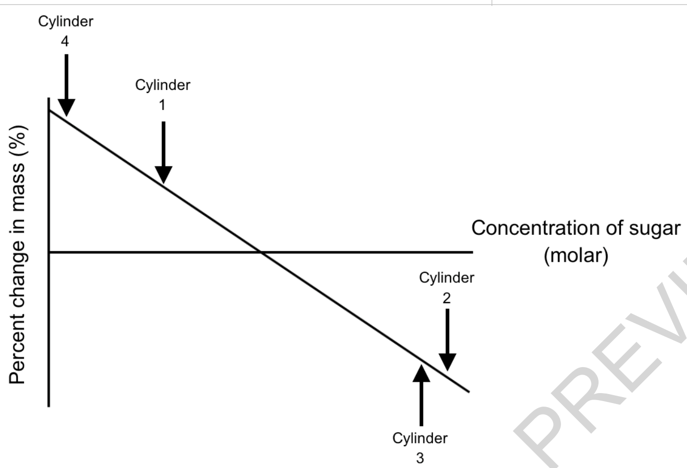


5d



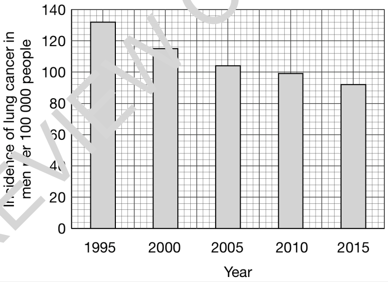
Where the line crosses the x axis, the potato cylinder has not changed mass. This means that there has been no overall movement of water in or out of the potato cylinder by osmosis. Because of this, we know that at this point the concentration of the solutions inside and outside of the potato cylinder are the same.

5e



Cardiovascular Diseases	
1a	Non-communicable diseases cannot be passed from one person to another by a pathogen. They are not infectious diseases.
1b	The coronary arteries branch out of the aorta and provide oxygen to the muscle cells of the heart.
1c	If the coronary arteries are blocked, not enough oxygen can be delivered in the blood to the heart muscles. This means that the heart may not be able to contract effectively.
1d	In coronary heart disease, fatty substances build up inside the coronary arteries. This causes the space (lumen) to narrow, restricting the amount of blood that can flow. The effect of this is to reduce the amount of oxygen that can be delivered to the heart muscle cells. If this becomes extreme, then the patient is suffering from a heart attack (where the heart muscle is starved of oxygen). A heart attack can be fatal.
1e	Statins reduce the level of certain types of cholesterol in the bloodstream. This slows down the rate that fatty materials build up in the arteries. Statins have been proven to reduce the risk of developing coronary heart disease. However, statins do have unwanted side effects for example they can cause liver problems.
1f	A stent can be used to treat people who have an almost total blockage of a coronary artery. A stent is a tube which is inserted into the coronary artery to keep it open. This allows the blood to flow normally through the artery.
1g	A stent does not treat the underlying causes of coronary heart disease so a person who has been fitted with a stent can still have narrowing of other regions of the coronary arteries.
2a	Heart valves do not fully open when the atria contract. In this case, the heart has to work extra hard to force the blood out of the atria and into the ventricles. This can make the heart become enlarged. Heart valves that leak blood when the ventricles contract. In this case, oxygenated blood is not pumped effectively to the body. Because of this, the patient may feel weak and tired.
2b	Mechanical valves: These can last a lifetime. Risk of blood clotting. Patient needs to take anti-clotting drugs for rest of life. Valves from animals: Do not last as long and may need to be replaced. Patients are not required to take drugs with this type of valve.
3a	In heart failure, the heart cannot pump enough blood around the body.
3b	Patients with heart failure may be treated with a heart transplant or with a heart-lung transplant. The problems here are: 1. There are not enough donated hearts / heart-lungs so there is a long waiting list. 2. The patient must take drugs to stop the donated heart from being rejected by the body's immune system. These drugs must be taken for the rest of the patient's life.
3c	Artificial hearts can only be used for a relatively short time because they increase the risk of blood clotting in the patient.

Required Practical: Photosynthesis	
1	Normal light bulbs emit a large amount of heat. This would increase the temperature of the pondweed and affect the rate of photosynthesis. An LED light source emits much less heat and is less likely to change the temperature of the pondweed.
2a	Sodium hydrogen carbonate solution is a source of carbon dioxide, which the plant needs for photosynthesis. If we simply used water instead, then the pondweed might not have enough carbon dioxide (in other words carbon dioxide may be a limiting factor). This means that the rate of photosynthesis would be extremely slow and we would not be able to investigate how the rate of photosynthesis is affected by light intensity.
2b	When the pondweed carries out photosynthesis, bubbles of oxygen gas will emerge from the cut end. By placing this at the top, it will be easier to count the bubbles. If we placed the cut end at the bottom, the bubbles could be difficult to count (for example passing behind the pondweed).
2c	When we place the pondweed into the solution, it will take the pondweed a few minutes to adjust to the conditions and start to photosynthesise. If we counted the bubbles of oxygen produced during this time, then we could get an inaccurate result. By waiting five minutes, we give the pondweed time to acclimatise to the conditions.
3a	The gas produced is oxygen which is a product of photosynthesis.
3b	The anomalous result is repeat 3 (12). This value is much lower than the other two values.
3c	To calculate the mean, we add up the numbers and then divide by the number of numbers. However, we must not include anomalous results when we calculate a mean. In the table, we have 36 and 40. This gives us a total of 76. Dividing 76 by 2 gives us a mean value of 38.
4	The species of pondweed = control variable as we must not change it during the experiment. Different species of pondweed will carry out photosynthesis at different rates. The concentration of sodium hydrogen carbonate solution = control variable. If we increase the concentration then the rate of photosynthesis could increase as more carbon dioxide is available. The volume of sodium hydrogen carbonate solution = control variable. If we increase the volume, then more carbon dioxide will be available and the rate of photosynthesis could increase. The distance between the LED light and the pondweed = independent variable. This is what we are varying in our experiment. The LED light source = control variable. If we used a different LED light source, the light intensity could change and this could affect the rate of photosynthesis. The temperature of the room = control variable. If the temperature of the room changed, this would affect the rate of photosynthesis. For example if the room temperature increased slightly, then the rate of photosynthesis could increase. We cannot control room temperature. However we could monitor it with a thermometer to check that it does not change. The number of bubbles produced per minute = dependent variable. This is what we measure for each change in the independent variable (in other words, this is what we measure to get our result).
5a	Problem 1: It is difficult to count the number of bubbles of oxygen accurately. This could be difficult when the light intensity is high (ie when the pondweed is close to the light source) as a lot of bubbles will be produced each minute. Problem 2: The bubbles of oxygen are different sizes. This means that a tiny bubble will be counted the same as a very large bubble.
5b	By using a funnel and upturned measuring cylinder, we can accurately measure the volume of oxygen produced in a given time. This overcomes the two problems mentioned above.
6a	Number of bubbles at 10 cm = 120, at 20 cm = 30, at 40cm = 7.
6b	The inverse square law means that when we increase the distance by x2, the number of bubbles per minute will decrease by x4 (ie 2^2). We can see this when we double the distance from 10 cm to 20 cm, the number of bubbles falls from 120 to 30 (ie 120 / 4). If we then double the distance from 20 cm to 40 cm, the number of bubbles falls to 7 (ie approximately 30 / 4).
6c	When we double the distance between the pondweed and the light source, the light intensity falls by a factor of 4x. Light is required for photosynthesis so this means that the rate of photosynthesis also falls by 4x.

Question	Answers	Mark	Extra information	Workbook page
1.1	A disease that cannot be passed from one person to another / not caused by a pathogen	1	Communicable diseases are spread by a pathogen.	Organisation page 66
1.2		1	It is important to be accurate when completing a graph or chart. There will be a small margin of error allowed but you need to be careful to be accurate.	Graph skills are not specifically addressed in any chapter but are covered throughout the book.
1.3	The data shown in Figure 1 are for men only. The data for women may be different.	1	Watch out for data which is specific to a group. It is not always possible to draw general conclusions from data such as this.	Organisation page 68
1.4	As the number of years of cigarette smoking increases, the risk of developing lung cancer also increases.	1	Be careful to link these two statements together. You would not get the mark if you simply stated that the risk of developing lung cancer increases.	Organisation page 68
1.5	A correlation does not prove a cause.	1	This is a very important idea in Biology. Correlation does not prove a cause. For example, people may start other habits around the same time that they start smoking (for example drinking alcohol). In order to prove that smoking causes lung cancer, we would need more evidence for example a causal mechanism.	Organisation page 68
1.6	If we sample a small number of people, then our results may show bias.	1	Sampling only provides reliable data if it is based on large numbers. If we sampled a small number of people, it is possible that these people do not represent the overall population. For example, they may be exposed to certain specific risk factors due to their occupation or where they live.	Organisation page 68
1.7	Carcinogens	1	Remember that carcinogens do not have to be chemicals. Ionising radiation is also a carcinogen.	Organisation page 69
1.8	Genes that control cell division change (accept mutate). Cell undergoes uncontrolled division. Malignant cancers spread to different parts of the body. Forming secondary tumours.	1 1 1 1	Remember that sometimes a cell undergoes uncontrolled division and forms a tumour but this does not spread. This is called a benign tumour. A benign tumour is not classed as a cancer.	Organisation page 64